

Acute Respiratory Distress Syndrome



Prepared by

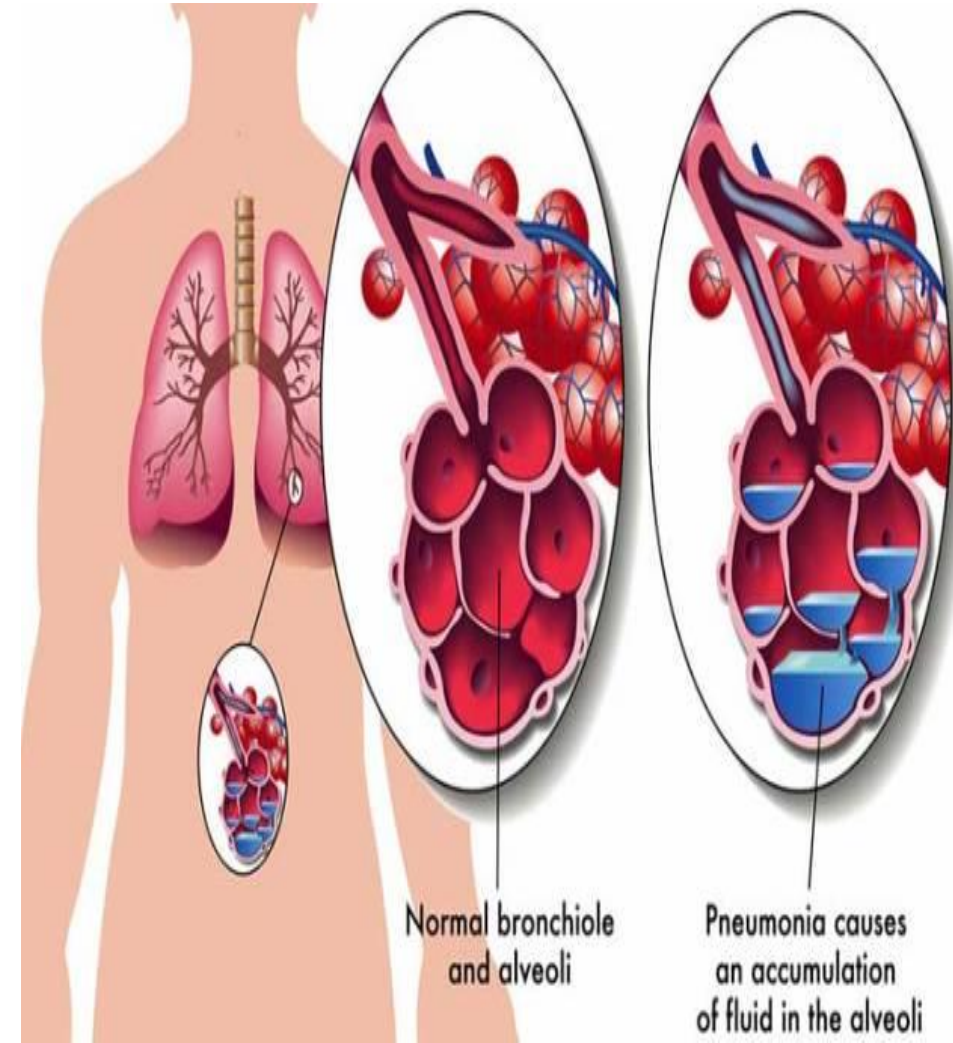
Assist professor Dr/Afaf Mohamed Mansour

outline

- Definition of ARDS
- Causes and risk factors for ARDS
- Pathophysiology
- Clinical diagnosis
- Medical management
- Nursing management

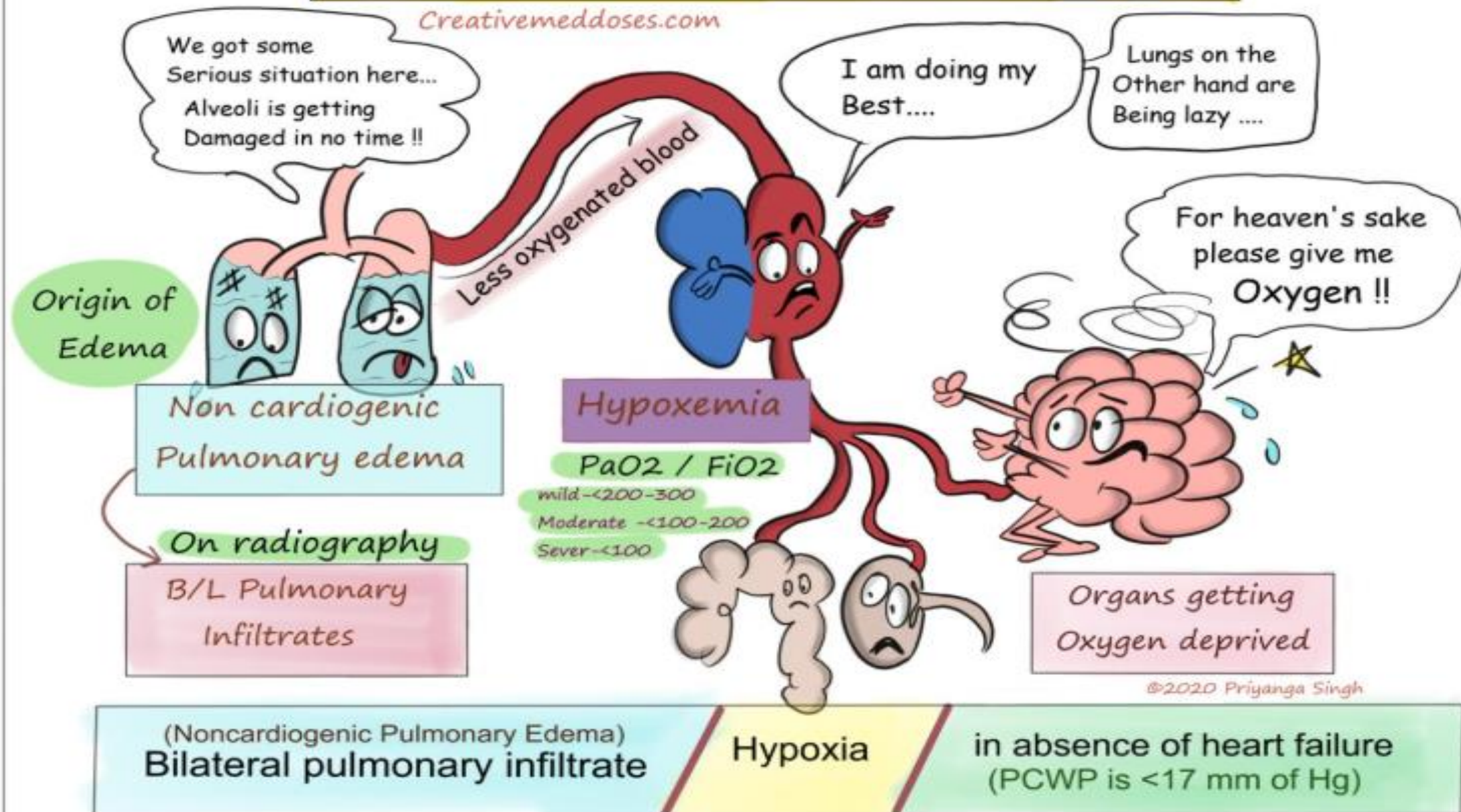
Definition of ARDS:

Acute respiratory distress syndrome (ARDS) is a life-threatening condition that occurs when there is severe fluid buildup in both lungs. The fluid buildup prevents the lungs from working properly which inhibit the transfer of oxygen from air into the body and carbon dioxide out of the body into the air.



ARDS: The Definition

Creativemeddoses.com



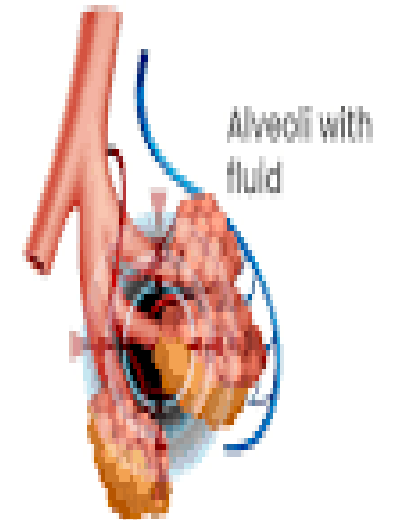
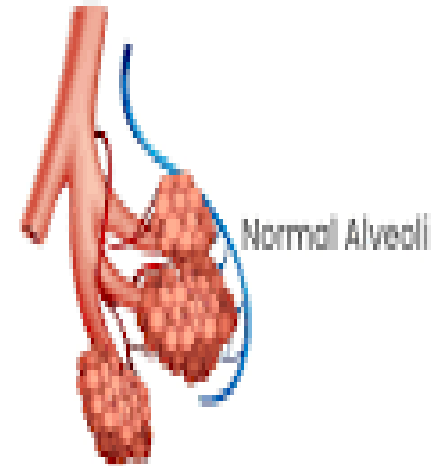
Causes and risk factors for ARDS:

Lung injury leading to ARDS can occur through

- ❑ A direct injury to the lungs
- ❑ Indirectly when a person is very sick or has a serious bodily injury

ARDS

Acute Respiratory Distress Syndrome



Direct Lung Injury

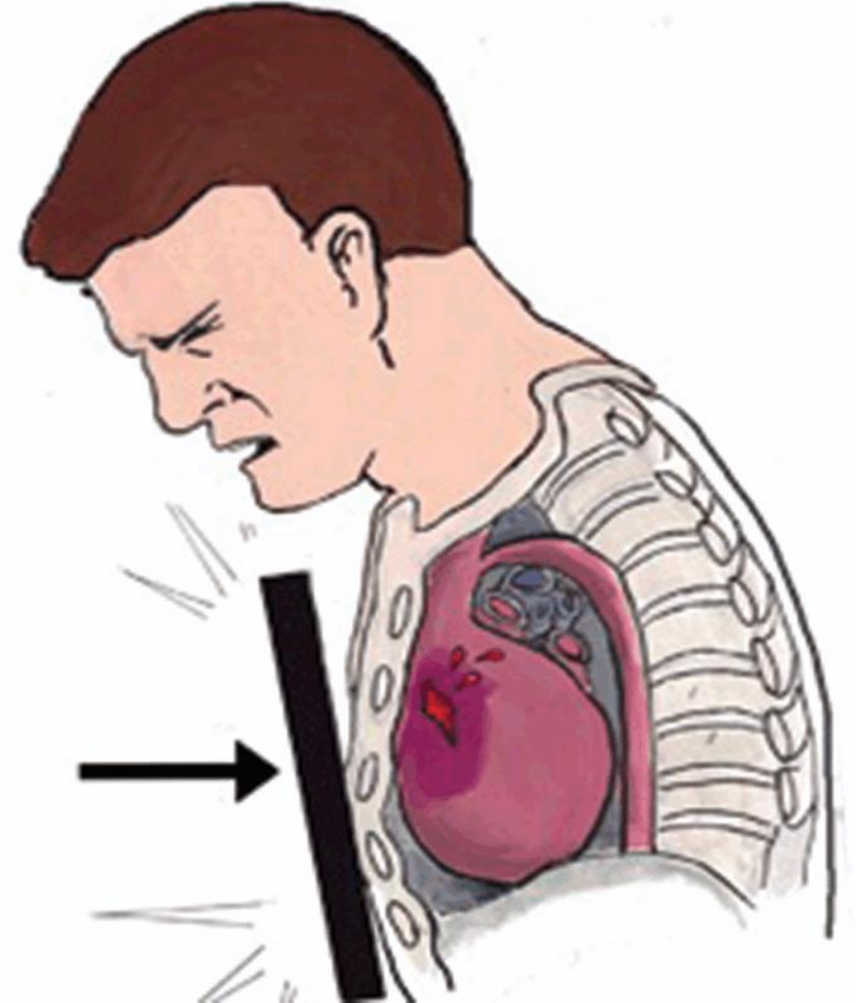
A direct injury to the lungs may result from breathing in harmful substances or an infection in the lungs. include:

- Severe viral , bacterial [pneumonia](#) and near drowning.
- **Coronavirus disease 2019 (COVID-19).**
- Breathing in or aspiration of gastric contents.
- Breathing in harmful fumes or smoke and toxic inhalations.



Direct Lung Injury

- A severe blow or injury to the chest from trauma (such as a car accident) or other accidents that causes bruising of the lungs.



Indirect Lung Injury

Most cases of ARDS happen in people who are very ill or who have been in a major accident.

- Severe injury or trauma with shock from a car crash, fire, or other cause.
- Severe and widespread bacterial infection in the body (sepsis)
- Severe bleeding requiring blood transfusions
- Drug reaction and drug overdose.

Indirect Lung Injury

- Acute pancreatitis.
- Systemic illnesses
- Long bone fracture.
- **General risk factors** including advanced age, female sex (noted only in trauma cases), cigarette smoking, and alcohol use and any severe illness can increase the risk of development of ARDS.

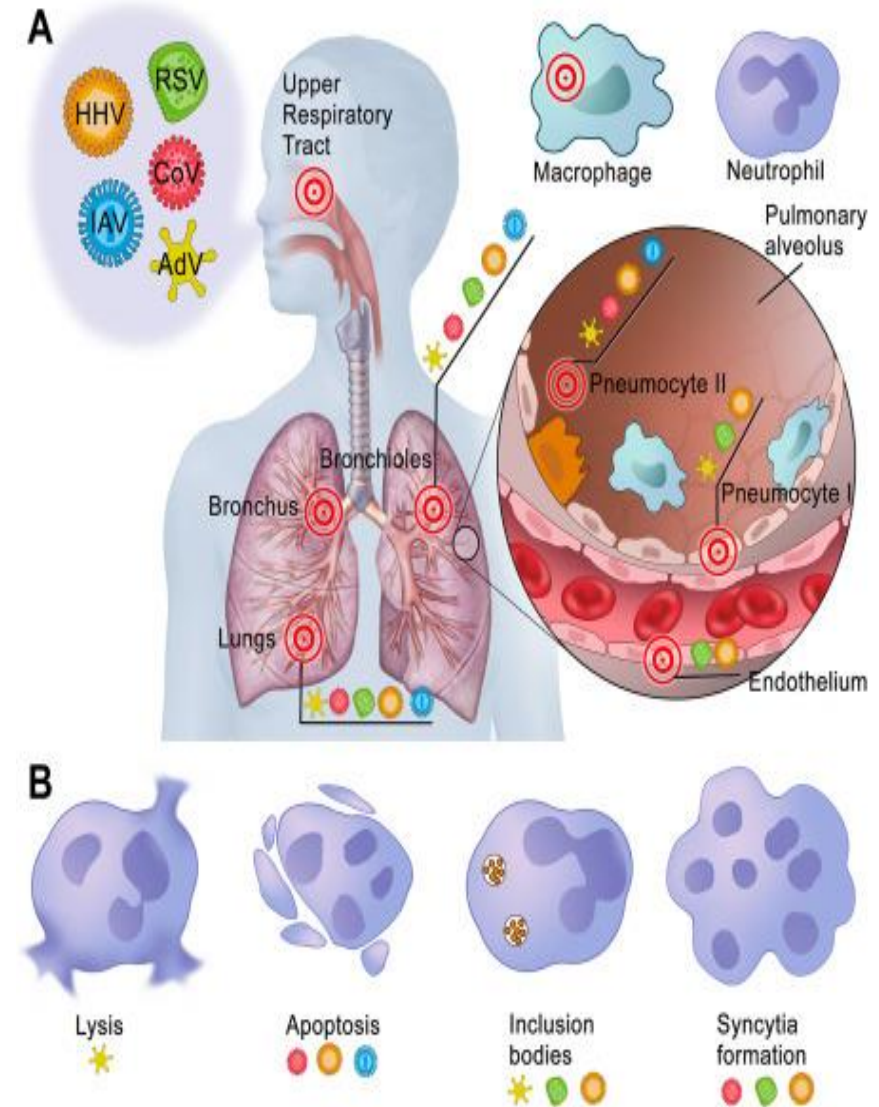
Pathophysiology

- Acute respiratory distress syndrome (ARDS) is associated with diffuse alveolar damage (DAD) and lung capillary endothelial injury. DAD is characterized by a diffuse inflammation of lung parenchyma.

usually results in an initial release of cytokines and other inflammatory mediators, secreted by local epithelial and endothelial cells

Pathophysiology

- Neutrophils and some T-lymphocytes quickly migrate into the inflamed lung parenchyma and contribute in the amplification of the phenomenon.
- Typical histological presentation involves diffuse alveolar damage and hyaline membrane formation in alveolar walls.

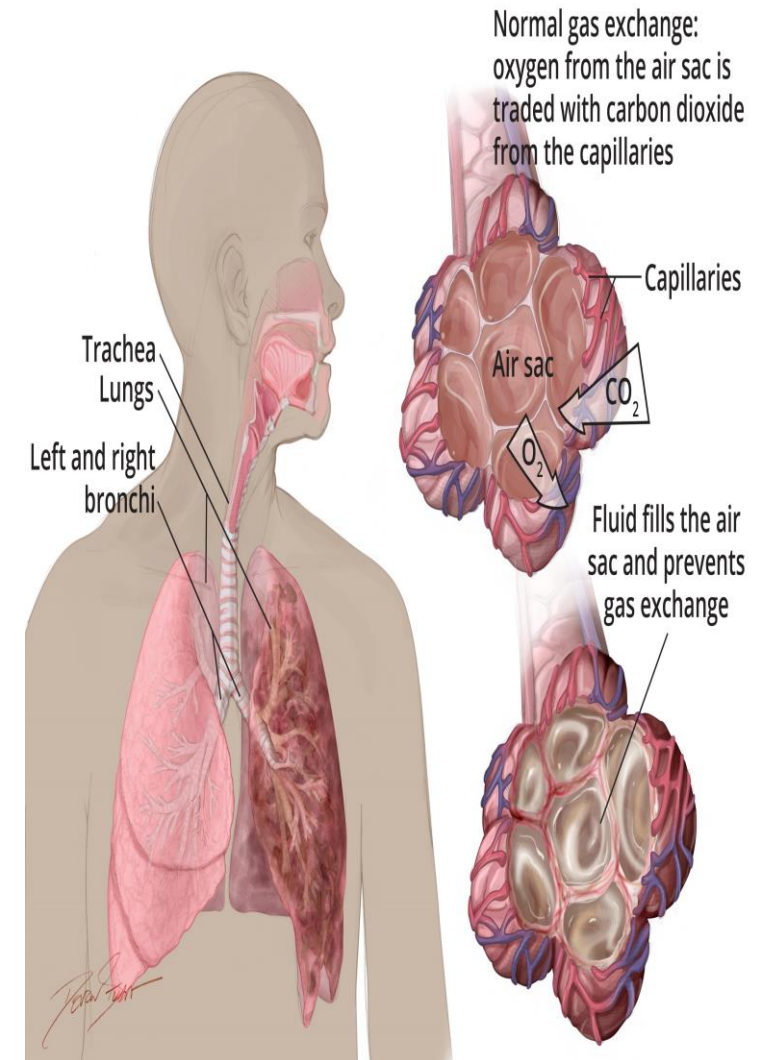


Pathophysiology

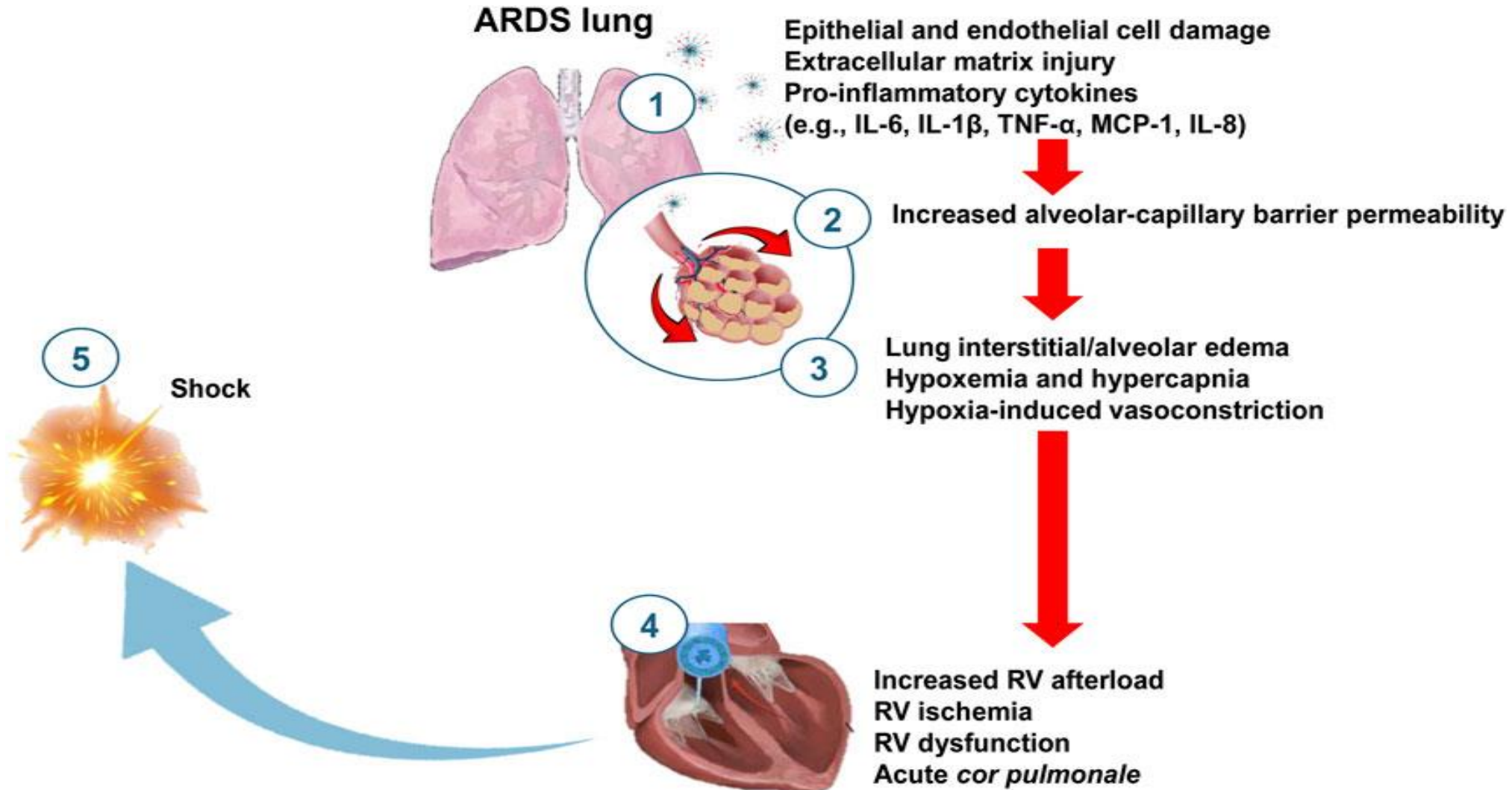
- ✓ The early phase is described as being exudative,
- ✓ The later phase is fibroproliferative in character.
- **Early ARDS** is characterized by an increase in the permeability of the alveolar-capillary barrier leading to an influx of fluid into the alveoli because of Inflammation alone, as in sepsis, causes endothelial dysfunction, fluid extravasation from the capillaries and impaired drainage of fluid from the lungs

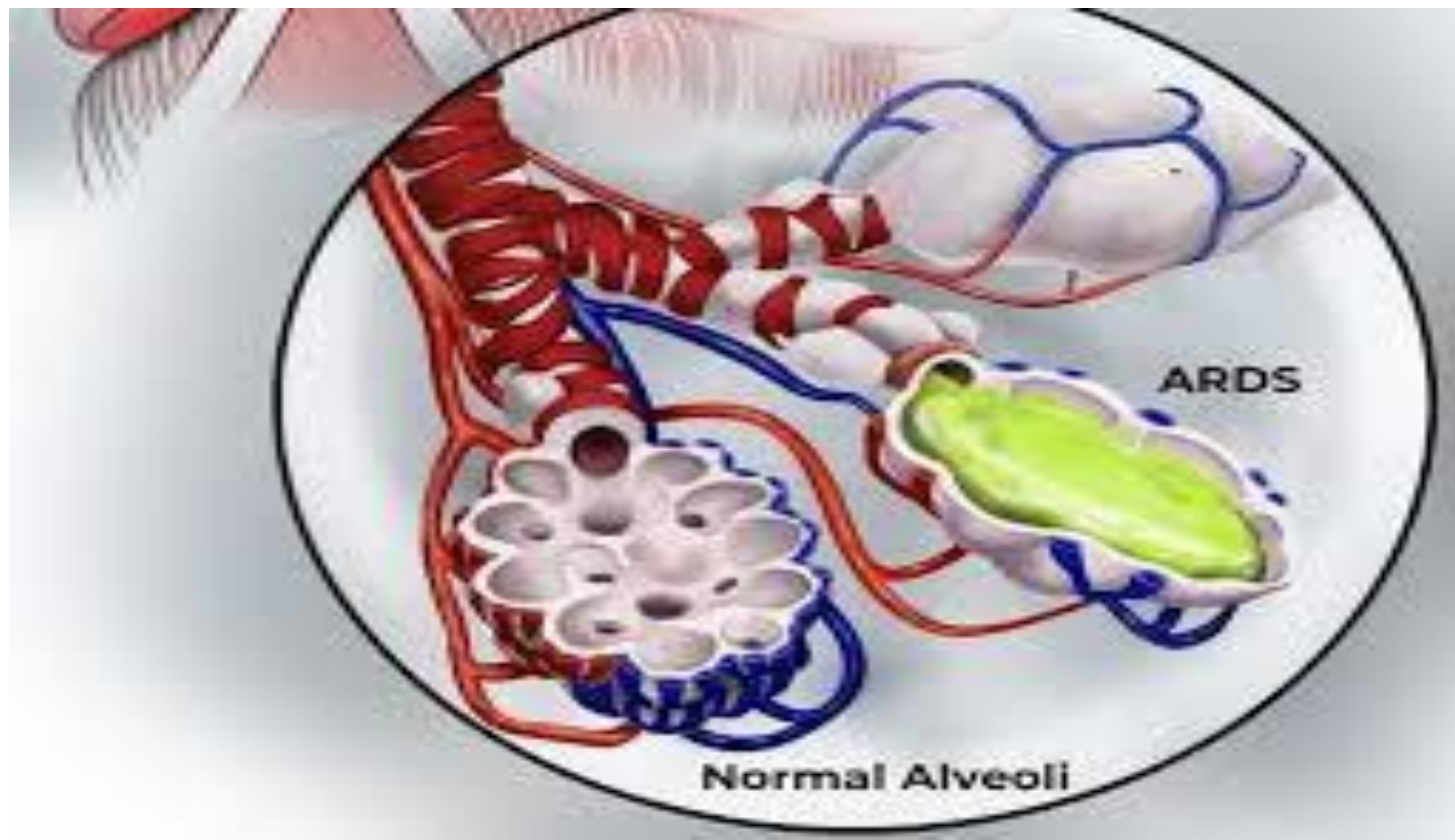
Pathophysiology

- **In a secondary phase**, endothelial dysfunction causes cells and inflammatory exudate to enter the alveoli. This pulmonary edema increases the thickness of the alveolo-capillary space, increasing the distance the oxygen must diffuse to reach blood. This impairs **gas exchange** leading to hypoxia, increases the work of breathing, eventually induces fibrosis of the airspace.



Pathophysiology





Clinical diagnosis

- ❖ **History**
- ❖ **Physical examination:**
- ❖ **A Plain chest x ray**
- ❖ **Blood tests**

Physical examination:

Include the major signs and symptoms of acute respiratory distress syndrome (ARDS) which are:

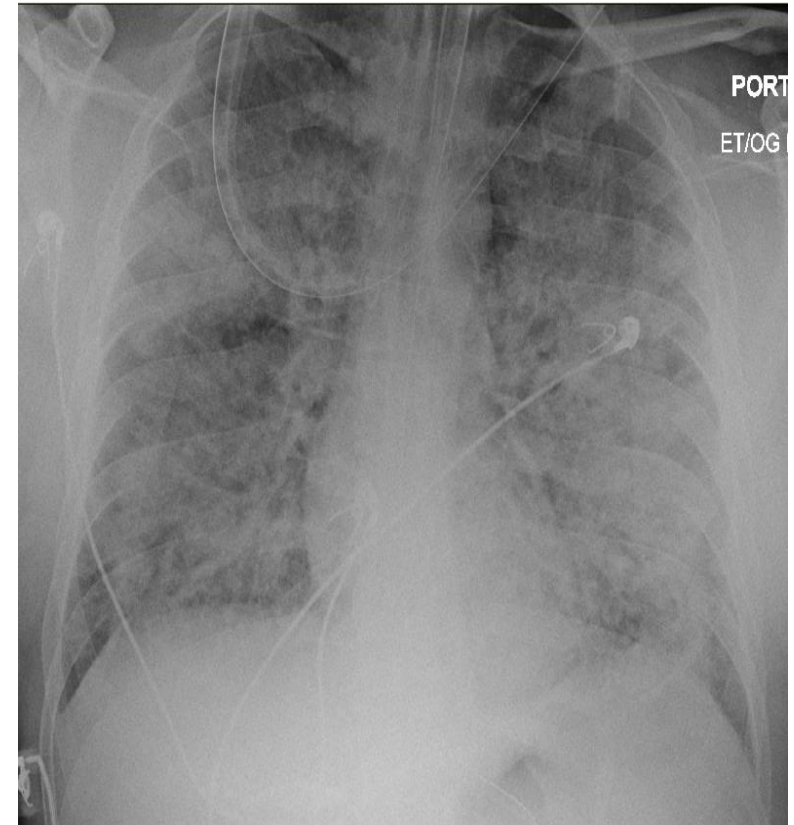
- . Shortness of breath
- . Fast, labored breathing

Physical examination:

- . Tachypnea, tachycardia, and the need for high inspired oxygen concentrations to maintain oxygen saturation.
- . Because acute respiratory distress syndrome (ARDS) often occurs because of sepsis, the patient may be **febrile or hypothermic** associated with hypotension and peripheral vasoconstriction with cold extremities may be present.
- . Cyanosis of the lips , nail beds and bluish skin color (due to a low level of oxygen in the blood).

Clinical diagnosis

- . **A Plain chest x ray** shows fluid in the air sacs of both lungs and bilateral alveolar infiltrates.
- . Because cardiogenic pulmonary edema must be distinguished from ARDS, carefully look for signs of **congestive heart failure or intravascular volume overload, including jugular venous distension, cardiac murmurs and gallops, hepatomegaly, and edema.**



Blood tests

- (ABG) Show a low level of oxygen in the blood
- Carefully examine sites of intravascular lines, surgical wounds, drain sites, and bed sores for evidence of infection.
- Check for subcutaneous air, a manifestation of infection or barotrauma.

Complications of ARDS:

- *Bacterial infections* The lungs or other parts of the body may become infected. These infections are usually treated with antibiotics after a culture and sensitivity test to detect the exact kind of bacteria is causing the infection

Air leaks.

- **Leaks of air** through holes in the lungs are caused by pressure from the breathing machine that is needed to be sure the patient gets enough air, and from the very stiff lungs.
- **Air** from the injured lungs may enter the pleura (the space between the lungs and the lining around the lungs) and cause a **pneumothorax** (collapsed lung). Treatment **using a chest tube and suction to remove the air and help the lungs reinflate.** **Air** may also enter the space between the membranes that line the abdomen (pneumoperitoneum) or the soft tissue under the skin (subcutaneous emphysema). These are not usually treated.

Other complications to be considered are:

. Pulmonary:

- Barotrauma (volutrauma)
- Pulmonary embolism (PE)
- Pulmonary fibrosis
- Ventilator-associated pneumonia (VAP)

. Gastrointestinal:

- Hemorrhage (ulcer)
 - Pneumoperitoneum
- Dysmotility
 - Bacterial translocation.

Other complications to be considered are:

- **Cardiac:** arrhythmias, myocardial dysfunction.
- **Renal:** acute renal failure (ARF), positive fluid balance.
- **Mechanical:** vascular injury, pneumothorax (by placing pulmonary artery catheter), tracheal injury/stenosis (result of intubation and/or irritation by endotracheal tube).
- **Nutritional:** malnutrition (catabolic state), electrolyte deficiency.

Medical management

- Bed rest
- Endotracheal intubation.
- **Mechanical ventilation** with positive end-expiratory pressure (PEEP) or continuous positive airway pressure (CPAP).
- Administer anesthetic to ease comfort during insertion of endotracheal tube: propofol
- Administer neuromuscular blocking agent—used when patients are on mechanical ventilation to avoid patient working against the action of the ventilator. These drugs allow respiratory muscles to rest: e.g. pancuronium, vecuronium

Medical management

- Administer **diuretics** to help decrease excess fluid in lungs: furosemide, ethacrynic acid, & bumetanide
- Administer **H2 blocker or proton pump inhibitor** to decrease gastric acid.
- This will decrease likelihood of a stress ulcer in the stomach or aspiration of gastric acid into the lung: e.g. ranitidine, famotidine, nizatidine, omeprazole
- Administer **anticoagulant**—clotting may have been causative in disease; immobility contributory to clot formation: • **heparin**

Medical management

- Administer **analgesic** for comfort and decrease myocardial oxygen demand: morphine
- Administer **steroids** to decrease inflammatory response in the lung tissue: hydrocortisone, methylprednisolone
- Administer **exogenous surfactant**: beractant
- Administer **antibiotics** for respiratory or systemic infections. Ideally selected based on results of culture and sensitivity (C&S) of sputum.

Nursing management

- Monitor WBC count:
 - Elevation of WBC with infection, inflammation.
 - Decrease in WBC in a patient who is immune-compromised or who has a viral infection.
- Monitor hemoglobin (Hgb) and hematocrit (Hct) for anemias.
- Monitor PT, PTT, and INR for coagulation abnormalities; monitor heparin dosing.

Nursing management

- Record intake and output of fluid:
- Monitor for signs of renal insufficiency or failure (decrease in urinary output less than 30 ml/h) and monitor BUN and Creatinine.
- Monitor for possible fluid overload—more fluid going in than coming out. Patient may end up in heart failure, compounding the fluid building up in the lungs.
- Weight the patient daily—inability to handle excess fluids, causing third spacing of fluids into interstitial spaces, increasing weight and causing edema.

Nursing management

- Change position every 2 hours to prevent pressure build-up, causing skin breakdown.
- Avoid overexerting the patient during treatment—patient will tire easily and will have problems with increased oxygen demands. Also provide rest periods during activities.
- Explain to the patient : The importance of doing coughing and deep-breathing exercises—after coming off the ventilator the patient needs to move adequate air in and out of the lungs. Coughing helps to rid

THANK YOU

A group of hands holding up large red letters spelling 'THANK YOU'. The letters are thick and three-dimensional. The hands are of various skin tones and are positioned at the bottom of the frame, holding the letters up. The background is plain white.